

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 September 2026
Team ID	SWTID-2026-5226
Project Name	Study Buddy API
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User & Profile Management	Register/login; create and update learner profile; secure authentication.
FR-2	Study Material Management	Create, upload, retrieve and organize notes, links and study resources.
FR-3	Study Planning & Tracking	Create study plans; set goals; track tasks, progress and completion status.
FR-4	Buddy Collaboration & Support	Find/connect with study buddies; share resources; send messages/updates; view buddy activity.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple REST API design, clear request/response formats and consistent error messages.
NFR-2	Security	Authentication, authorization, password protection and validation of user input.
NFR-3	Reliability	Consistent data handling, validation and graceful error recovery for API operations.
NFR-4	Performance	Efficient API responses and database queries for common study and collaboration actions.
NFR-5	Availability	Service should remain accessible with appropriate error handling and health monitoring.
NFR-6	Scalability	API architecture should support growth in users, study resources, plans and buddy interactions.

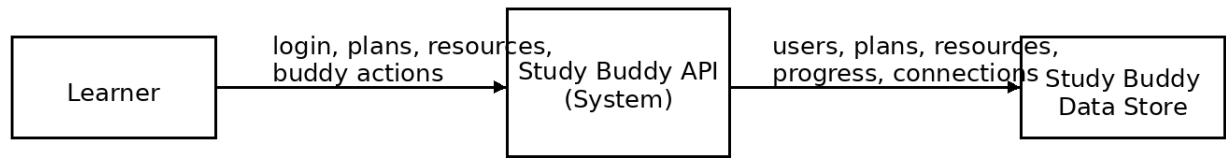
Study Buddy API – User Stories & Data Flow Diagrams

User Stories

Story	User Story
US-01	As a learner, I want to register and log in securely so that I can access my study data.
US-02	As a learner, I want to create and organize study resources so that I can find materials quickly.
US-03	As a learner, I want to create study plans and tasks so that I can manage my preparation.
US-04	As a learner, I want to track task completion and progress so that I can monitor my study goals.
US-05	As a learner, I want to find and connect with study buddies so that I can collaborate and stay accountable.
US-06	As a learner, I want to share study resources and updates with my buddies so that we can learn together.

Data Flow Diagrams

DFD Level 0 - Study Buddy API



DFD Level 1 - Study Buddy API

